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| | **AIR UNIVERSITY** |
| | **DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING** |
| | **EXPERIMENT NO 1** |

**Lab Title:** Introduction to Arrays

**Group Members:** Muhammad Waqas **Reg. No:** 240756

**Objective:** Learn what arrays are and how they work in C++.Practice how to create, use, and solve problems using 1D and 2D arrays.

**LAB ASSESSMENT:**

| Attributes | Excellent (5) | Good (4) | Average (3) | Satisfactory (2) | Unsatisfactory (1) |
|---|---|---|---|---|---|
| Ability to Conduct Experiment | | | | | |
| Ability to assimilate the results | | | | | |
| Effective use of lab equipment and follows the lab safety rules | | | | | |

Total Marks: _____ Obtained Marks: _____

**LAB REPORT ASSESSMENT:**

| Attributes | Excellent (5) | Good (4) | Average (3) | Satisfactory (2) | Unsatisfactory (1) |
|---|---|---|---|---|---|
| Data presentation | | | | | |
| Experimental results | | | | | |
| Conclusion | | | | | |

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____




**Department of Computer Engineering**

**Data Structures And Algorithms**

**LAB REPORT : 2**

**Tittle : Introduction To Arrays**

**Name : Muhammad Waqas**

**Roll No: 240756**

**Submitted To : CE . Engr Maryam**

**Date : 27 September 2025**

**Question 1 :**

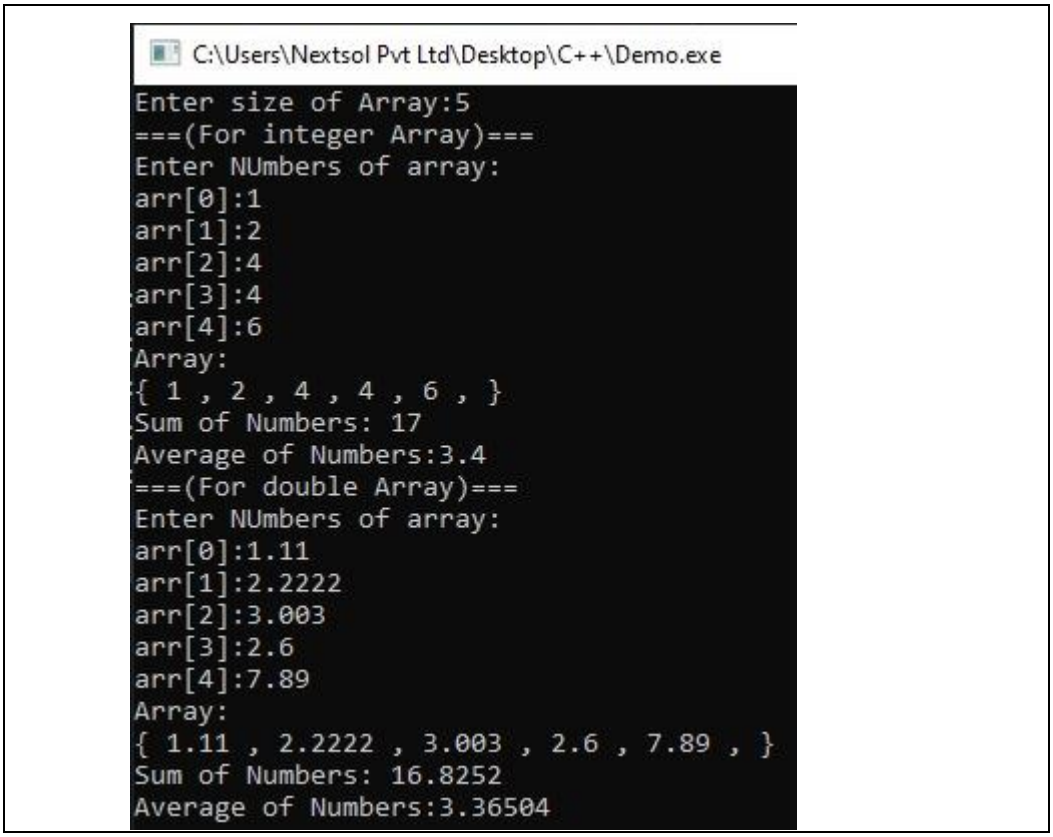
**Calculate Sum and Average of N Numbers**

**With class**

most.cpp > Display_Data()

```cpp
#include<iostream>
using namespace std;
template<typename T>
class Most_class{
private:
T* arr;
int n;
public:
Most_class(int n1);
~Most_class();
void Enter_Data();
void Display_Data();
void repeated_var(); };
template<typename T>
Most_class<T>::Most_class(int n1){
n=n1;
arr=new T[n];
for(int i=0;i<n;i++){
arr[i]=0;}}
template<typename T>
Most_class<T>::~Most_class(){
delete[] arr;
cout<<"Destructor\n";}
template<typename T>
void Most_class<T>::Enter_Data(){
cout<<"Enter array elements:\n";
for(int i=0;i<n;i++){
cout<<"a["<<i<<"]: ";cin>>arr[i];}}
template<typename T>
void Most_class<T>::Display_Data(){
cout<<"{";
for(int i=0;i<n;i++){
 cout<<"Array:\n";
cout<<"{ ";
for(int i=0;i<n;i++){
cout<<arr[i]<<" , "; }
 cout<<"}";}
template< typename T>
void Array<T>::average(){
double sum=0;
for(int i=0;i<n;i++){
sum+=arr[i];    }
double avg=static_cast<double>(sum)/n;
cout<<"\nSum of Numbers: "<<sum<<endl;
cout<<"Average of Numbers:"<<avg<<endl;}
int main(){
int size1;
cout<<"Enter size of Array:";
cin>>size1;
 Array<int> s(size1); //For int array
Array<double> f1(size1); //For double arr

cout<<"===(For integer Array)===\n";
s.EnterData();
s.display_data();
s.average();
cout<<"===(For double Array)===\n";
 f1.EnterData();
 f1.display_data();
 f1.average();
system("pause");
cout<<"End";
 return 0;}
```


C:\Users\Nextsol Pvt Ltd\Desktop\C++\Demo.exe
Enter size of Array:5
===(For integer Array)===
Enter NUmbers of array:
arr[0]:1
arr[1]:2
arr[2]:4
arr[3]:4
arr[4]:6
Array:
{ 1 , 2 , 4 , 4 , 6 , }
Sum of Numbers: 17
Average of Numbers:3.4
===(For double Array)===
Enter NUmbers of array:
arr[0]:1.11
arr[1]:2.2222
arr[2]:3.003
arr[3]:2.6
arr[4]:7.89
Array:
{ 1.11 , 2.2222 , 3.003 , 2.6 , 7.89 , }
Sum of Numbers: 16.8252
Average of Numbers:3.36504

Question 2 :

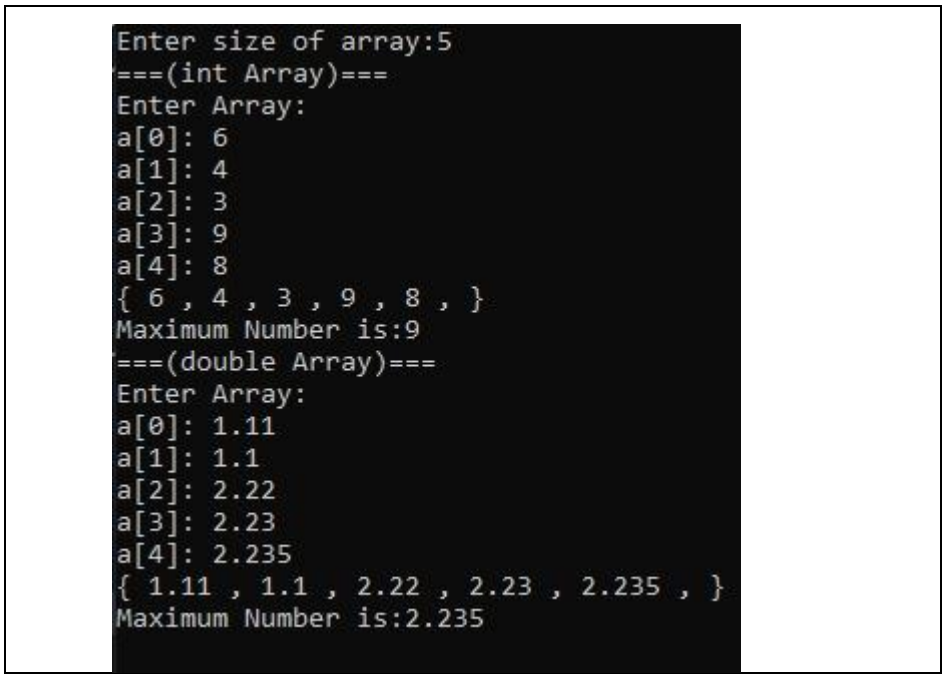
Write a C++ program to find the largest element of a given array of integers (using a class, no vector).

most.cpp > Display_Data()

```cpp
#include<iostream>
using namespace std;
template<typename T>
class Most_class{
private:
T* arr;
int n;
public:
Most_class(int n1);
~Most_class();
void Enter_Data();
void Display_Data();
void repeated_var(); };
template<typename T>
Most_class<T>::Most_class(int n1){
n=n1;
arr=new T[n];
for(int i=0;i<n;i++){
arr[i]=0;}}
template<typename T>
Most_class<T>::~Most_class(){
delete[] arr;
cout<<"Destructor\n";}
template<typename T>
void Most_class<T>::Enter_Data(){
cout<<"Enter array elements:\n";
for(int i=0;i<n;i++){
cout<<"a["<<i<<"]: ";cin>>arr[i];}}
template<typename T>
void Most_class<T>::Display_Data(){
cout<<"{";
for(int i=0;i<n;i++){
```

```cpp
void Max_class<T>::display_data(){
cout<<"{ ";
for(int i=0;i<n;i++){
cout<<arr[i]<<" , ";}
cout<<"}\n";}
template< typename T>
void Max_class<T>::max_number(){
T maxnum=arr[0];
for(int i=0;i<n;i++){
if(maxnum<arr[i]){
maxnum=arr[i];}}
cout<<"Maximum Number is:"<<maxnum<<endl;}
int main(){
int size1;
cout<<"Enter size of array:";
cin>>size1;
Max_class<int> a1(size1);
Max_class<double> b1(size1);
cout<<"===(int Array)===\n";
a1.Enter_data();
a1.display_data();
a1.max_number();
cout<<"===(double Array)===\n";
b1.Enter_data();
b1.display_data();
b1.max_number();
return 0;}
```

**Output:**

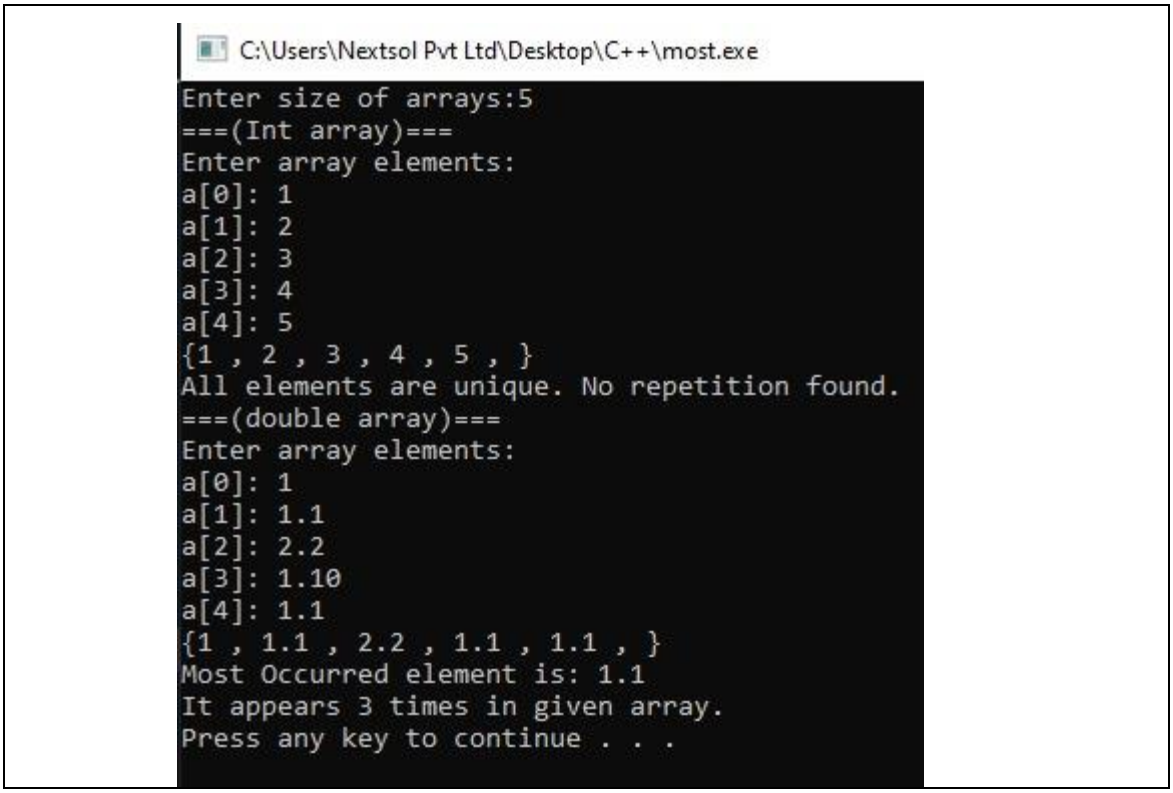

Enter size of array:5
===(int Array)===
Enter Array:
a[0]: 6
a[1]: 4
a[2]: 3
a[3]: 9
a[4]: 8
{ 6 , 4 , 3 , 9 , 8 , }
Maximum Number is:9
===(double Array)===
Enter Array:
a[0]: 1.11
a[1]: 1.1
a[2]: 2.22
a[3]: 2.23
a[4]: 2.235
{ 1.11 , 1.1 , 2.22 , 2.23 , 2.235 , }
Maximum Number is:2.235

**Question 3 :**

**Write a C++ program to find the most occurring element in an array of integers (using a class, no vector).**

most.cpp > Display_Data()

```cpp
#include<iostream>
using namespace std;
template<typename T>
class Most_class{
private:
T* arr;
int n;
public:
Most_class(int n1);
~Most_class();
void Enter_Data();
void Display_Data();
void repeated_var(); };
template<typename T>
Most_class<T>::Most_class(int n1){
n=n1;
arr=new T[n];
for(int i=0;i<n;i++){
arr[i]=0;}}
template<typename T>
Most_class<T>::~Most_class(){
delete[] arr;
cout<<"Destructor\n";}
template<typename T>
void Most_class<T>::Enter_Data(){
cout<<"Enter array elements:\n";
for(int i=0;i<n;i++){
cout<<"a["<<i<<"]: ";cin>>arr[i];}}
template<typename T>
void Most_class<T>::Display_Data(){
cout<<"{";
for(int i=0;i<n;i++){
for(int i=0;i<n;i++){
cout<<arr[i]<<" , ";}
cout<<"}\n";}
template<typename T>
void Most_class<T>::repeated_var(){
int maxcount = 0;
T most_var = arr[0];
for(int i=0;i<n;i++){ int count=0;
for(int j=0;j<n;j++){
if(arr[i]==arr[j]){count++;}}
if(count>maxcount){
maxcount=count;
most_var=arr[i];}}
if(maxcount == 1) {
cout << "All elements are unique.";
} else {
cout << "Most Occurred element is: "
<< most_var << endl;
cout << "It appears " << maxcount
<< " times in given array.\n";}}
int main(){ int size;
cout<<"Enter size of arrays:";cin>>size;
Most_class<int> m1(size);
Most_class<double> m2(size);
cout<<"===(Int array)===\n";
m1.Enter_Data(); m1.Display_Data();
m1.repeated_var();
cout<<"===(double array)===\n";
m2.Enter_Data();m2.Display_Data();
m2.repeated_var();
system("pause");return 0;}
```


C:\Users\Nextsol Pvt Ltd\Desktop\C++\most.exe
Enter size of arrays:5
===(Int array)===
Enter array elements:
a[0]: 1
a[1]: 2
a[2]: 3
a[3]: 4
a[4]: 5
{1 , 2 , 3 , 4 , 5 , }
All elements are unique. No repetition found.
===(double array)===
Enter array elements:
a[0]: 1
a[1]: 1.1
a[2]: 2.2
a[3]: 1.10
a[4]: 1.1
{1 , 1.1 , 2.2 , 1.1 , 1.1 , }
Most Occurred element is: 1.1
It appears 3 times in given array.
Press any key to continue . . .

## Conclusion:

- Arrays store multiple values in a single variable using a fixed size.
- Pointers can be used to access and manipulate elements in an array.
- The name of an array acts like a pointer to its first element.
- Using pointers with arrays helps in dynamic memory handling and efficient looping.
- Changes made using pointers directly affect the original array elements.
- Understanding arrays with pointers is important for memory management in DSA.